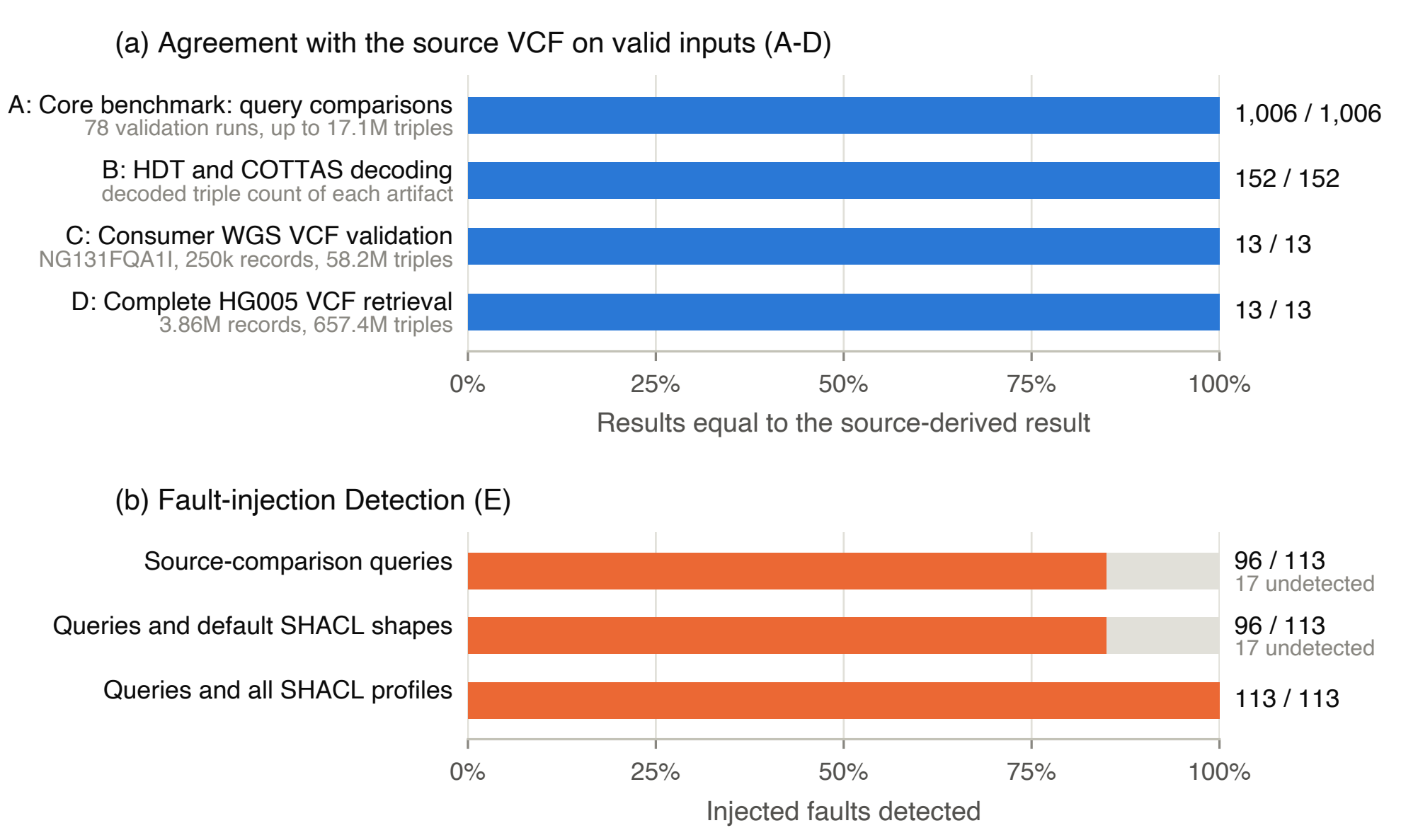
(a) Agreement with the source VCF on valid inputs (A-D)
A: Core benchmark: query comparisons
78 validation runs, up to 17.1M triples
1,006 / 1,006
B: HDT and COTTAS decoding
decoded triple count of each artifact
152 / 152
C: Consumer WGS VCF validation
NG131FQA1I, 250k records, 58.2M triples
13 / 13
D: Complete HG005 VCF retrieval
3.86M records, 657.4M triples
13 / 13
0%
25%
50%
75%
100%
Results equal to the source-derived result
(b) Fault-injection Detection (E)
Source-comparison queries
96 / 113
17 undetected
Queries and default SHACL shapes
96 / 113
17 undetected
Queries and all SHACL profiles
113 / 113
0%
25%
50%
75%
100%
Injected faults detected